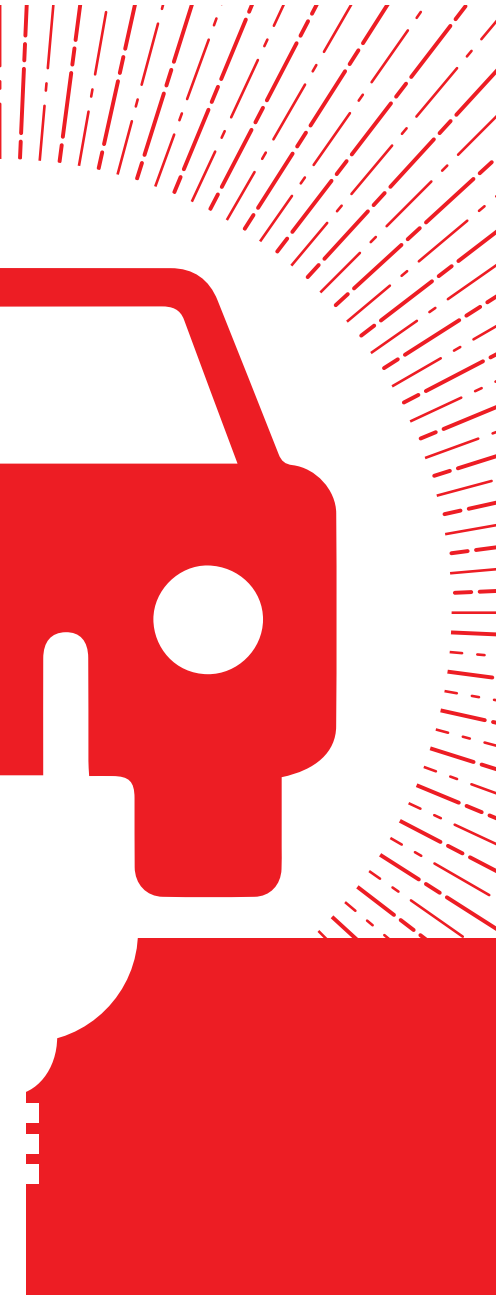


cent since 2019, those of the E-variants have grown over 100 per cent. Although the E-rickshaws account for close to 90 per cent of the total E-variants and purists tend to turn their noses up at the ungainly contraptions, they effectively meet the same need. And as they move in tandem with consumer expectations to higher horsepower motors, li-ion batteries, hydraulic braking systems and better body structures, they have ensured that 100 per cent electrification before 2030 seems quite attainable.

ELECTRIC PASSENGER CARS COULD WELL **STAY RESTRICTED TO PRODUCTS AT THE HIGHER END** OF THE PRICE SPECTRUM IN INDIA

Cars—Battery Electric Vehicles (BEV)—unsurprisingly, have tended to lag. Given the low base, the growth numbers look impressive at first glance. At just over 30,000 units sold in 2022, BEV passenger cars represent just about 1 per cent of the total car registrations in India last year. A large number of these with electric powertrains shoehorned into existing ICE variants have not quite attained globally expected levels of finesse in power delivery or even user interface, but they represent a significant indigenous capability in electrical/ electronic architecture development. What is less certain is whether the present level of product pricing—helped significantly by fiscal support—is economically sustainable at present levels. Quick back-of-the-envelope calculations suggest that some of the leaders in that segment could be leaving a fair amount of red ink—at least Rs 1.25 lakh per unit—on their manufacturers' books. I'm of course assuming no financial callisthenics, and high ESG compliance standards.

At the heart of the pricing matter is the battery. Even if we look at lower capacity packages, batteries will remain fully dependent on a globally constrained supply chain. Look, for example, at what Zimbabwe has just decided—a ban on the export of unprocessed lithium ores and an effective control by Chinese conglomerates of both lithium mining and ore-beneficiation in that country. This trend is only expected to significantly speed up and could well cover cobalt as well. Will an Organisation of Lithium Exporting Countries (OLEC) become a new-technology OPEC? And as demand for their raw materials continues to be driven upwards by EV introduction plans in richer countries, the cartelisation of lithium producers might only work to push up prices, thus crowding out the mass consumer demand in countries like India simply based on price. And neither can swapping provide a viable solution for cars. Swapping a 12-kg E2W battery is very different from trying to swap even a 120-kg small car battery. True, technology developments like solid state and new chemistry might help lower the size of the battery, but no one is committing any real progress on that front before 2026. Going forward, BEV passenger cars in India could, therefore, well stay restricted to products at the higher end of the price spectrum. Hydrogen Fuel Cell EVs (FCEV) and even strong hybrids could well prove the most effective solutions for the Indian passenger car market.

As for buses, no move could be strategically more ill-advised than the reckless push for BEV buses, making a mockery of the prime minister's exhortations for *atmanirbharta*, and this effectively puts the throats of our urban transport system comprehensively in the clutches of the 'Dragon'! Converting existing city bus fleets to 'trolley buses' and operating with fully indigenous urban power grids could be a simpler, more intelligent and cost-effective solution. The government announcement on a National Hydrogen Mission, therefore, comes not a day too soon. Hopefully, some part of the research outlays will find their way to developing a vastly distributed production of hydrogen from biomass, which can then meet the needs of Hydrogen FCEV buses at lower costs. One estimate suggests that by using just 200 million tonnes of unutilised agricultural residues, India can produce up to 5 million tonnes of truly 'green' hydrogen annually, which can meet the requirements of at least half the intercity commercial vehicle fleets. ■

Illustration by **TANMOY CHAKRABORTY**